```c
//C Program: Multiplication Table Using while Loop

#include <stdio.h>

int main()

{

    int num, i = 1;

    printf("Enter a number: ");

    scanf("%d", &num);

    while (i <= 10)

    {

        printf("%d x %d = %d\n", num, i, num * i);

        i++;

    }

   return 0;

}
```

Output:

```
Enter a number: 5

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35
```

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

2. // Pyramid Pattern using Nested for loop

```
#include <stdio.h>

int main()

// Pyramid Pattern

{

   int i, j, space;

   for(i = 1; i <= 5; i++)

   {

      for(space = 1; space <= 5 - i; space++)

      {

         printf(" ");

      }

      for(j = 1; j <= (2 * i - 1); j++)

      {

         printf("*");

      }

      printf("\n");

   }

   return 0;

}
```

Output:

```
    *
   ***
  *****
 *******
*********
```

Pyramid Pattern

Logic

This program uses **three loops**:

1. **Outer loop (i)** → Controls the rows.
2. **First inner loop (space)** → Prints spaces before the stars to center the pyramid.
3. **Second inner loop (j)** → Prints stars using the formula (2 × i − 1).

Step-by-Step

- Row 1 → 4 spaces, 1 star.
- Row 2 → 3 spaces, 3 stars.
- Row 3 → 2 spaces, 5 stars.
- Row 4 → 1 space, 7 stars.
- Row 5 → 0 spaces, 9 stars.

Formula

- **Spaces** = 5 – i
- **Stars** = 2 × i - 1

3. // Number Triangle using do while loops

```
#include <stdio.h>

int main()

{

   int i = 1, j;

   do
```

```
    {
        j = 1;  // j is a reset value
        do
        {
            printf(" %d", j);
            j++;
        } while (j <= i);
        printf("\n");
        i++;
    } while (i <= 5);
    return 0;
}
```

Output:

```
 1
 1 2
 1 2 3
 1 2 3 4
 1 2 3 4 5
```